

DEM'YANSKOYE, Russian Federation

WMO: 280760

Lat: 59.5921N

Lon: 69.3445E

Elev: 187

StdP: 14.60

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.2	-28.9	-39.6	0.6	-34.0	-34.0	0.8	-28.7	11.4	-6.5	9.8	-4.6	1.8	80	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	83.2	67.9	80.3	66.7	76.9	65.1	70.4	79.5	68.4	76.9	66.6	74.5	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	100.8	74.9	65.4	94.5	72.8	63.6	88.5	70.9	34.5	79.5	32.8	77.0	31.4	74.7	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.8	9.8	8.7	DB	-36.6	88.8	7.3	1.7	-41.8	90.0	-46.1	91.0	-50.2	92.0	-55.5	93.2
(5)				WB	-37.1	73.6	7.2	1.7	-42.3	74.8	-46.5	75.8	-50.5	76.8	-55.7	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	-2.7	2.4	19.7	33.6	47.5	60.0	63.7	57.9	47.8	35.0	16.3	4.5
	DBStd	25.37	15.38	14.26	11.25	10.48	10.18	8.73	7.08	7.30	7.49	9.41	14.00	15.42
	HDD50	7626	1633	1332	938	497	174	16	1	12	128	472	1012	1411
	HDD65	12081	2098	1752	1403	943	548	197	113	242	515	932	1462	1876
	CDD50	1171	0	0	0	4	96	317	427	258	64	5	0	0
	CDD65	153	0	0	0	0	6	49	74	23	1	0	0	0
	CDH74	1348	0	0	0	0	96	446	635	166	5	0	0	0
	CDH80	274	0	0	0	0	19	94	135	26	0	0	0	0
Wind	WSAvg	3.7	3.0	3.2	3.8	4.5	4.7	4.1	3.8	3.6	3.6	3.7	3.7	3.1
Precipitation	PrecAvg	23.0	1.1	0.9	1.1	1.1	1.9	2.6	3.0	3.5	2.6	1.9	1.7	1.5
	PrecMax	32.4	2.0	1.8	2.8	2.6	3.5	4.6	10.8	6.1	5.1	4.9	2.8	2.8
	PrecMin	16.4	0.0	0.1	0.0	0.1	0.3	1.2	0.9	1.4	0.5	0.1	0.2	0.6
	PrecStd	3.5	0.6	0.4	0.6	0.6	0.8	1.0	1.9	1.5	1.1	1.0	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	34.9	45.9	65.3	82.8	86.7	86.9	83.2	73.6	62.0	42.3	34.2
		MCWB	31.5	32.7	38.0	52.3	61.1	69.2	70.4	69.8	62.4	51.9	40.2	33.2
	2%	DB	27.1	31.3	40.2	57.2	76.5	82.7	83.5	78.6	67.5	56.2	38.0	31.3
		MCWB	26.4	29.7	33.9	47.7	59.3	67.2	68.9	67.4	59.1	49.3	36.8	30.5
	5%	DB	22.7	26.1	36.7	52.7	71.5	79.5	80.9	74.5	63.1	51.4	34.9	28.5
		MCWB	21.8	24.3	32.0	44.4	56.7	65.8	67.8	65.5	56.5	47.1	33.9	27.7
	10%	DB	18.9	22.0	34.0	48.5	65.6	75.5	77.4	70.6	59.2	47.7	32.5	24.3
		MCWB	18.1	20.3	30.8	41.6	54.0	64.2	66.2	63.1	54.4	44.6	31.4	23.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.8	33.6	38.7	53.2	64.0	72.0	73.4	71.8	63.9	53.2	40.4	33.6
		MCDB	32.1	34.8	45.2	62.1	77.3	83.2	83.2	81.1	70.1	59.2	42.1	34.2
	2%	WB	26.4	29.6	35.0	49.3	60.4	69.1	71.2	68.7	60.0	50.2	37.0	30.9
		MCDB	27.1	31.4	38.8	56.3	73.7	79.1	80.4	76.1	65.5	55.2	37.9	31.5
	5%	WB	22.1	24.4	33.0	45.1	57.7	67.2	69.2	66.5	57.4	47.5	34.2	27.7
		MCDB	22.8	25.9	35.8	51.3	69.5	76.6	77.7	73.0	62.1	51.1	34.8	28.4
	10%	WB	18.1	20.4	31.1	42.0	54.8	65.2	67.4	64.1	54.8	44.8	31.7	23.6
		MCDB	18.8	22.0	33.8	48.2	64.7	73.7	75.1	69.4	58.7	47.7	32.4	24.4
Mean Daily Temperature Range	5% DB	MDBR	12.9	16.3	17.6	16.7	19.3	17.8	17.5	16.2	13.9	10.6	11.1	12.3
		MCDBR	14.4	14.5	17.1	21.5	27.2	23.5	22.7	21.6	19.2	16.6	8.5	11.0
		MCWBR	14.1	13.0	12.2	13.5	13.5	11.0	10.2	11.5	11.5	10.9	8.1	10.4
	5% WB	MCDBR	14.0	13.5	14.4	20.2	25.0	21.1	20.8	20.0	18.0	15.9	8.5	11.3
		MCWBR	14.0	12.5	11.2	13.6	13.6	11.0	10.4	11.2	11.5	10.8	8.3	11.1
Clear-Sky Solar Irradiance	taub	0.269	0.308	0.365	0.395	0.398	0.405	0.416	0.420	0.365	0.336	0.290	0.259	
	taud	2.000	2.012	1.942	2.066	2.273	2.327	2.295	2.246	2.368	2.274	2.077	1.946	
	Ebn at Noon	152	210	232	252	263	263	256	243	240	204	142	106	
	Edh at Noon	16	28	41	43	38	36	37	36	27	22	15	11	
All-Sky Solar Radiation	RadAvg	162	453	929	1361	1583	1742	1674	1204	760	355	143	86	
	RadStd	11	34	61	146	130	127	158	96	76	45	27	10	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page